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Salvadoran student working across applied mathematics, quantitative risk, machine learning and systems: original research in tail-risk forecasting and cryptography, a national AI hackathon win, and automation tools used in real accounting work. I make the design decisions and build with AI agents.

EDUCATION

Complejo Educativo "Thomas Jefferson" El Salvador
Technical Baccalaureate in Technology Infrastructure and IT Services *Expected 2026*

- Straight-10 grades and first place in class every year.
- Coursework: networks, operating systems, information security and programming.
- Conducted an ISO 27001 audit of a savings and credit cooperative: risk analysis, access-control and policy review, and a gap report.

EXPERIENCE

Accounting Assistant, Process Automation Feb 2026 – Present
El Salvador

- Built Partidas DTE, a desktop app that reads identifiers from accounting entries (PDF) and merges the matching electronic invoices into one file, cutting a task that took a full afternoon (~4 hours) to under 1 minute (over 99% faster), recurring monthly across multiple clients.
- Built DTE Joiner, a second tool that reads the purchases ledger (Excel) and assembles the matching invoice PDFs in ledger order, with resumable progress, missing-invoice alerts and an audit log; similar reduction (~4 hours to under 1 minute).
- Developed Excel/VBA macros that generate payroll sheets and journal entries with automatic balance checks, saving an estimated 30 to 40% of payroll time.
- Automated extraction, cleaning and validation of financial data with Python.
- Contributed to the implementation of an ERP for a maquila manufacturer (inventory, invoicing, production control, reports).

Math Tutor, Community Service 200 hours

- Taught mathematics classes, including to grades above my own.

RESEARCH & PROJECTS

CryptoRisk github.com/mariomarroquin3/cryptorisk

- Compared 20+ volatility and tail-risk models (GARCH family, HAR, CAViaR, MS-GARCH, RF-QR, LSTM) for 1-day VaR and Expected Shortfall on BTC and ETH with a 500-day walk-forward protocol; evaluated with Kupiec, Christoffersen, Dynamic Quantile, Acerbi-Szekely, FZO, Diebold-Mariano and MCS tests.
- Deployed a live dashboard (Next.js, React, Recharts) at cryptorisk-mauve.vercel.app.
- Wrote the methodology in LaTeX; findings: Realized-GARCH and CAViaR lead the FZO ranking, MS-GARCH fails to identify stable regimes with a 500-day window.

Entropy Arena github.com/mariomarroquin3/entropy-arena

- Designed a framework to compare randomness sources for cryptographic key generation (BIP39/32/48/67, 2-of-3 and 3-of-5 multisig), separating statistical quality (NIST SP 800-22) from real entropy (NIST SP 800-90B); recovered the MT19937 state in 0.02 s and RANDU in 0.26 s.

Violence, Institutions and Growth github.com/mariomarroquin3/data_analyzer

- Built a reproducible two-way fixed-effects pipeline on 8 countries (2000–2024) with small-cluster inference (wild cluster bootstrap, CR2, Driscoll-Kraay), diagnostics and ML triangulation.

Grainy · Coffee quality prediction National AI Hackathon winner (2025)

- Built a Streamlit app that predicts coffee quality with Random Forest regressor and classifier models, generates PDF reports and includes a Gemini chatbot (RAG with FAISS) over a technical guide.
- Now rebuilding it with leak-free validation: a "pre-cupping" model without sensory scores, de-duplicated datasets and validation by origin.

RetinoVisionAI EUREKA challenge

- Trained an EfficientNet-B3 classifier for diabetic retinopathy with augmentation, illumination normalization and stratified cross-validation; added Grad-CAM interpretability and a web app (PyTorch, Flask).

Financial data synthesizer (VAE) github.com/mariomarroquin3/data_generator

- Built a Variational Autoencoder (PyTorch) that generates synthetic financial transactions for testing accounting software without exposing real data, with a Streamlit app and PCA validation (Code in Place final project).

RFID access control (IoT) github.com/mariomarroquin3/logger

- Implemented an ESP32 + RFID system that validates cards against Firebase Realtime Database, drives entry/exit barriers with servo motors and logs every event; React/TypeScript dashboard with live statistics and Web Serial credential registration.

Data center monitoring (team project) github.com/mariomarroquin3/modulo3.3

- Set up Zabbix on an Ubuntu virtual machine to collect metrics from classmates' computers; co-built a React report analyzing an interview with a Tier 4 data-center engineer under ISO 22237 and ISO 22301.

ERP for a maquila (TypeScript) github.com/mariomarroquin3/ERP_maquila

- Built an ERP with inventory, invoicing, production control and reports as a technical baccalaureate project.

Data Miner · multi-agent team (BoxLang AI Hackathon 2026) github.com/mariomarroquin3/dataMiner

- Team of four; I built the agent architecture and backend: coordinator plus three agents, personal-data anonymization, read-only SQL/MCP and re-execution of the SQL behind each figure; extracted the module github.com/mariomarroquin3/bx-datatools.

Synapse · Bank database (school team project) github.com/mariomarroquin3/synapse_1

- Designed the Access database around a double-entry ledger (symmetric accounting entries) and wrote queries and services in Python/pyodbc: atomic transfers, account-status rules, cards, roles; Streamlit; team of five.

HONORS, LEADERSHIP & PROGRAMS

- Winner, first National Artificial Intelligence Hackathon (2025): coffee quality classification with a Random Forest model.
- Multiple wins in district and departmental public speaking contests.
- Mathematics olympiads, two consecutive years.
- Member, Cohere Labs Open Science Community and Google Developer Program; participant, ML Summer School 2026.
- In progress: GCI Global 2026 credit-risk competition (Matsuo Lab, University of Tokyo); Kaggle Enveda CASMI 2026; CUBO+ and CUBO_ai; Stanford Probability for AI.
- AWS ML Scholars program completed.

CERTIFICATIONS

- AWS: ML Engineer Associate Curriculum Overview; Introduction to Machine Learning; SimuLearn Cloud Computing Fundamentals (2026).
- Stanford Code in Place (2026, Python).
- Cisco Introduction to Data Science (2026).
- Udacity AWS AI Practitioner Challenge (2026).
- Platzi: Linear Algebra for ML; Math Functions for Data Science and AI; Math Language and Notation; Math Fundamentals (2026).
- Google Developer Program badges: ML Crash Course (Linear and Logistic Regression), Gemini Enterprise Agent Ready, Google Cloud & NVIDIA community member, Chrome DevTools, DOM Detective, Android Studio.
- Daxus LATAM: Power BI + AI Career Accelerator, 8 h (2026).
- Direct English, Intermediate (2024).

SKILLS & LANGUAGES

Programming: Python, SQL, LaTeX. Working knowledge with AI assistance: TypeScript, JavaScript/React, Java, VBA; learning R.
ML & Data: PyTorch, scikit-learn, statsmodels, arch, Pandas, NumPy, Matplotlib, Streamlit, Flask.
Analytics: Power BI, SQL, Excel/VBA, Plotly, Altair.
Infrastructure & IoT: AWS (training), Zabbix, Ubuntu/Linux, virtual machines, Firebase, ESP32, RFID, Vercel, Render.
Web: Next.js, React, TypeScript, Flask, Git/GitHub.
AI-assisted engineering: architecture and feature design with AI agents and MCP.
Areas: Time series, econometrics, financial risk, cryptography, linear algebra, statistics, deep learning.
Languages: Spanish (native); English (intermediate certificate 2024, advanced level in preparation).